Capstone Project-I Report

On

Real-Time Sign Language Recognition System

submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING



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May, 2026

SUPERVISOR CERTIFICATE

This is to certify that the project report entitled "Real-Time Sign Language Recognition System Using MediaPipe and XGBoost" submitted by Dev Dixit (2K23CSUN06003), Harshita Maggo (2K23CSUN06007) and Raghav Gaur (2K23CSUN06016) is a bona fide record of the work carried out by them under my supervision and guidance.

This work has not been submitted elsewhere for the award of any other degree/diploma. The project fulfills the requirements of the course Capstone Project-I as prescribed by Manav Rachna University.

To the best of my knowledge, the work presented in this report is original and has been carried out in accordance with the academic standards and guidelines.

Date:

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STUDENT DECLARATION

We, Dev Dixit (2K23CSUN06003), Harshita Maggo (2K23CSUN06007) and Raghav Gaur (2K23CSUN06016) are hereby to declare that the project entitled "Real-Time Sign Language Recognition System " is a record of original work carried out by me under the guidance of Dr. Neha Chaudhary in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Artificial Intelligence and Robotics from Manav Rachna University.

I further declare that this project has not been submitted previously, in part or in full, for the award of any other degree, diploma, or certification in this or any other institution.I also affirm that all sources of information used in this project have been properly acknowledged and cited in the report.

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ABSTRACT

Sign language is the primary mode of communication for millions of individuals with hearing and speech impairments worldwide. However, the lack of widespread sign language literacy among the general population creates a significant communication barrier, leading to social exclusion and reduced accessibility in everyday interactions. This project addresses this critical gap by developing a real-time sign language recognition system that leverages computer vision and machine learning to bridge the communication divide.

The proposed system employs Google's MediaPipe Hands framework to extract 21 three-dimensional hand landmarks from live webcam input, generating a 63-feature vector (x, y, z coordinates per landmark) for each captured frame. These features are fed into an XGBoost classifier, replacing the traditional Random Forest approach, to achieve superior gesture recognition accuracy. The classifier is trained on a custom dataset of 3,316 samples spanning five gesture classes: YES, NO, HELLO, BATHROOM, and NONE (neutral), collected using an automated data collection pipeline.The system achieves a test accuracy of 97.44% with per-class F1-scores exceeding 0.95 across all gesture categories. A real-time inference pipeline processes webcam frames at approximately 10 frames per second, maintaining low latency on standard CPU hardware without requiring GPU acceleration. The system incorporates a rolling-buffer sentence builder that accumulates stable gesture predictions into meaningful sentences, and a text-to-speech module using gTTS that vocalizes the built sentence on user command.

The complete application is deployed as an interactive web interface using Gradio, hosted on HuggingFace Spaces, making it accessible without local installation. Confidence scores for each gesture class are visualized directly on the video feed using OpenCV overlays, providing transparency in the model's decision-making process.

The system architecture is modular, allowing straightforward extension to additional gesture classes and potential integration of larger ASL datasets from platforms such as Kaggle for enhanced vocabulary coverage.

This project demonstrates a practical, lightweight, and deployable AI solution for assistive communication technology, with direct implications for accessibility, inclusive design, and human-computer interaction research.

Keywords: Sign Language Recognition, MediaPipe, XGBoost, Hand Landmark Detection, Gesture Classification, Real-Time Inference, Gradio, Assistive Technology, Text-to-Speech

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1. Introduction

1.1 Background

Sign language is a complete, natural language that uses visual-manual modality to convey meaning, employing combinations of hand shapes, orientation, movement, and facial expressions. Globally, over 70 million people are deaf, and hundreds of millions more have hearing impairments, relying on sign language as their primary communication medium. The absence of effective, real-time translation tools between sign language and spoken/written language creates an acute accessibility gap in healthcare, education, employment, and public services.[2]

Recent advances in computer vision, particularly deep learning-based pose estimation frameworks such as MediaPipe, have enabled real-time extraction of hand skeletal structures from standard webcam feeds. Combined with classical machine learning classifiers and modern gradient boosting algorithms, these frameworks enable the construction of accurate, low-latency gesture recognition systems that can run entirely on commodity CPU hardware.[3]

1.2 Problem Statement

The communication barrier between sign language users and the general population remains largely unaddressed by existing commercial technology. Current solutions are either expensive, require specialized hardware, rely on heavy deep learning models unsuitable for real-time deployment, or lack the accuracy needed for practical use. A lightweight, accessible, and accurate real-time sign language recognition system deployable via a standard web browser is therefore a critical unmet need in assistive technology.

1.3 Motivation

The motivation for this project stems from the observable disparity in communication accessibility for hearing-impaired individuals. Despite the availability of machine learning tools and computer vision libraries, practical deployable solutions that non-

technical users can access remain rare. This project is motivated by the potential to democratize sign language recognition, making it available through a simple browser interface without requiring any specialized hardware or software installation on the user spart . Additionally, as a student researcher with experience in applied machine learning and robotics, the development of this system represents a tangible contribution to assistive AI, aligning with broader goals of inclusive technology design.

1.4 Scope

The current scope of the project encompasses the following:

- Recognition of five American Sign Language (ASL)-inspired gestures: YES, NO, HELLO, BATHROOM, and NONE
- Real-time gesture detection from standard webcam input using MediaPipe Hands
- Classification using XGBoost trained on a custom dataset of 3,316 samples
- Sentence building from stable gesture sequences with configurable cooldown and confidence thresholds
- Text-to-speech conversion of built sentences using gTTS
- Web-based deployment via Gradio on HuggingFace Spaces

Out of scope for the current version: full ASL alphabet recognition, mobile deployment, multi-hand gesture support, and facial expression integration.

1.5 Objectives

The primary objectives of this project are:

- To develop a real-time hand gesture recognition pipeline using MediaPipe landmark extraction and XGBoost classification
- To achieve a minimum test accuracy of 95% on the custom gesture dataset
- To implement a sentence builder that converts stable gesture predictions into coherent text
- To integrate text-to-speech output for complete gesture-to-speech conversion
- To deploy the system as a publicly accessible web application on HuggingFace Spaces

- To maintain real-time performance (minimum 8 FPS inference) on standard CPU hardware without GPU dependency

1.6 Applications

- Assistive communication tools for deaf and hard-of-hearing individuals in healthcare and public service interactions
- Educational tools for teaching sign language to hearing individuals
- Human-computer interaction systems enabling gesture-based control
- Integration into video conferencing platforms for automatic sign language captioning
- Accessibility features in smart devices and kiosks
- Research baseline for larger-scale sign language recognition systems

1.7 Mapping with Relevant SDGs

This project aligns with the following United Nations Sustainable Development Goals (SDGs):

- SDG 3 (Good Health and Well-Being): Improves healthcare communication accessibility for deaf patients, reducing diagnostic errors due to communication barriers.
- SDG 4 (Quality Education): Enables inclusive education by providing a tool that bridges sign language and spoken language in academic environments.
- SDG 10 (Reduced Inequalities): Directly reduces communication inequality experienced by hearing-impaired communities by providing accessible, free-to-use technology.
- SDG 17 (Partnerships for the Goals): Leverages open-source frameworks (MediaPipe, XGBoost, Gradio) and public deployment platforms (HuggingFace) to maximize accessibility and collaboration.

2. Literature Review and Comparative Analysis

Sign language recognition has been an active research area for over two decades. Early systems relied on data gloves with embedded sensors to capture hand joint angles, offering high accuracy but requiring expensive, uncomfortable hardware that limited practical adoption.[19] The emergence of computer vision-based approaches, particularly following the availability of high-quality RGB depth cameras and the release of the Microsoft Kinect, shifted research toward camera-based gesture recognition. [13]

Convolutional Neural Networks (CNNs) demonstrated strong performance on static gesture recognition from image input, with models such as ResNet-50 and MobileNet achieving high accuracy on standard ASL datasets.[9] However, these approaches require significant training data, GPU hardware for real-time inference, and are susceptible to background variation and lighting changes. Recurrent architectures including Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRUs) have been applied to dynamic gesture sequences with promising results, but at the cost of significantly higher computational demands. [13]

The introduction of Google's MediaPipe framework marked a significant advancement, enabling real- time 3D hand landmark extraction at 21 keypoints on standard CPU hardware. Several studies have leveraged MediaPipe landmarks as input features for lightweight classifiers. Amarjot Singh et al. (2021) demonstrated that Random Forest classifiers trained on normalized MediaPipe landmarks could achieve above 90% accuracy on static ASL signs with minimal computational overhead. Subsequent work by Taskiran et al. (2022) explored gradient boosting methods for the same task, showing improvements in minority class performance and confidence calibration.[1][3]

The present work builds on this foundation by adopting XGBoost as the classifier of choice, motivated by its superior handling of imbalanced class distributions (relevant

given the varying sample counts across our gesture classes), built-in probability estimation for confidence visualization, and competitive inference speed. A comparative summary of relevant approaches is presented below:[12]

Approach	Input Type	Accuracy	GPU Required	Real-Time
CNN (ResNet-50)	Full Image	~96%	Yes	Limited
LSTM + CNN	Video Sequence	~94%	Yes	No
Data Glove + SVM	Sensor Data	~98%	No	Yes
MediaPipe + Random Forest	Landmarks	~93%	No	Yes
MediaPipe + XGBoost (Ours)	Landmarks	97.44%	No	Yes

Table 2.1: Comparative analysis of sign language recognition approaches

The proposed system outperforms the traditional MediaPipe + Random Forest baseline by 4.44 percentage points in overall accuracy, while maintaining CPU-only real-time inference capability. The XGBoost model additionally provides calibrated probability outputs, enabling the confidence visualization and stability-based sentence building features of the system.[2][20]

3. Proposed Solution

3.1 Methodology

The system follows a modular pipeline architecture consisting of four sequential stages: (1) video capture and preprocessing, (2) hand landmark extraction, (3) gesture classification, and (4) post- processing for sentence building and speech synthesis. This design ensures that each component can be independently updated or replaced without affecting the rest of the pipeline.[5][4]

Video frames are captured from the system webcam via OpenCV and passed to the MediaPipe Hands solution, which detects and tracks up to one hand per frame. The detected landmarks are normalized and flattened into a 63-dimensional feature vector. This vector is passed to the trained XGBoost classifier, which returns both a class prediction and a probability distribution across all five gesture classes. The prediction and confidence scores are overlaid on the video frame using OpenCV drawing functions, and the prediction is passed to the sentence builder for accumulation into a coherent sentence.[4]

3.2 System Design

The system comprises the following key modules:

predictor.py: Encapsulates the MediaPipe Hands session and XGBoost model. Accepts an RGB frame, extracts landmarks, runs inference, draws confidence overlays on the frame, and returns the annotated frame along with prediction metadata.

sentence_builder.py: Maintains a rolling deque buffer of the last 25 frame predictions. A gesture is added to the sentence when a single class constitutes at least 72% of the buffer, confidence exceeds 60%, and a 2-second cooldown has elapsed since the last addition.

tts_engine.py: Converts the built sentence to speech using gTTS, saving the output as a temporary MP3 file returned to the Gradio audio component for browser

playback.app.py: Constructs the Gradio Blocks interface with a streaming webcam input, annotated video output, sentence textbox, status indicator, and Speak/Clear/Remove-last control buttons. The .stream() method drives the real-time inference loop at 10 FPS.

3.3 Hardware / Software Requirements

Hardware Requirements:

- Standard webcam (720p or higher recommended)
- CPU: Intel Core i5 or equivalent (no GPU required)
- RAM: Minimum 4 GB
- Internet connection for gTTS and HuggingFace Spaces deployment

Software Requirements:

- Python 3.10+
- MediaPipe 0.10.14
- XGBoost 2.0+
- Gradio 4.28+
- OpenCV (opencv-python-headless 4.9+)
- scikit-learn 1.3+, joblib 1.3+
- gTTS 2.3.2+
- Conda virtual environment (recommended)

3.4 Algorithms / Pseudocode

Algorithm 1: Real-Time Gesture Inference

```
INPUT: Webcam frame (RGB) OUTPUT: Prediction label, confidence, annotated
frame
1. Pass frame to MediaPipe Hands 2. IF no hand detected: return frame,
'none',
0.0 3. Extract 21 landmarks → flatten to 63-dim vector 4. Run
XGBoost.predict_proba(vector) 5. pred_idx = argmax(probabilities) 6.
prediction
= classes[pred_idx] 7. confidence = probabilities[pred_idx] 8. Draw
prediction label and confidence bars on frame 9. RETURN
annotated_frame, prediction, confidence
```

Algorithm 2: Sentence Builder

```
INPUT: prediction (string), confidence (float) OUTPUT: Boolean (new word
added)
1. IF confidence < 0.60: append 'none' to buffer
2. ELSE: append prediction to buffer
3. IF 'none' in effective: reset last_gesture
4. IF buffer not full: RETURN False
5. dominant = mode(buffer)
6. stability = count(dominant) / buffer_size
7. IF stability >= 0.72 AND dominant != last_gestureAND (now -
last_time) >= 2.0:      sentence.append(dominant) last_gesture =
dominant; last_time = nowRETURN True
8. RETURN False
```

3.5 Dataset Description

The training dataset was collected using a custom Python script (Gesture_Ai_data.py) that leverages MediaPipe to extract hand landmarks from live webcam input and saves the corresponding 63-feature vectors to CSV files, one per gesture class. Data collection was performed under varied lighting conditions and hand positions to improve generalization.[4]

Gesture Class	Samples Collected	Test F1-Score
YES	1049	0.99
BATHROOM	713	0.96
NO	688	0.97
NONE	495	0.95
HELLO	371	0.99
TOTAL	**3316**	**0.97 (avg)**

Table 3.1: Dataset composition and per-class model performance

Features: 63 numerical values representing the normalized x, y, z coordinates of 21 hand landmarks as extracted by MediaPipe Hands. No image data is stored; only the landmark coordinates are persisted, minimizing storage requirements and eliminating privacy concerns associated with raw image datasets.

3.6 Initial Prototype Demonstration

The initial prototype is fully functional and accessible at http://localhost:7860 during local execution, and will be publicly available at the project's HuggingFace Spaces URL upon deployment. The prototype demonstrates:

- Live webcam feed with MediaPipe hand skeleton overlay (green landmarks and white connections)[3]
- Top-left prediction label displaying current gesture and confidence percentage

- Bottom-left confidence bar visualization showing probability distribution across all five classes
- Real-time sentence accumulation displayed in the Sentence textbox below the video feed
- Voice output via the Speak button, converting the built sentence to audio
- Control buttons for removing the last word and clearing the sentence[6]

The system correctly identifies all five gesture classes with high confidence under standard indoor lighting conditions, with the XGBoost model achieving 97.44% test accuracy on held-out data.

3.7 Achievement of Objectives as Per Plan

Objective	Target	
Gesture recognition pipeline (MediaPipe + XGBoost)	>95% accuracy	Achieved (97.44%)
Real-time inference on CPU	<200ms latency	Achieved (~100ms)
Sentence builder with stability buffer	Functional	Achieved
Text-to-speech integration	Functional	Achieved
Gradio web interface	Functional	Achieved
HuggingFace Spaces deployment	Public URL	In Progress

Table 3.2: Objective achievement status

4. Project Planning

4.1 Gantt Chart

The project was executed over an 8-week development cycle, organized into the following phases:

Phase	W1	W2	W3	W4	W5	W6	W7	W8
Literature Review	■	■						
Data Collection	■	■	■					
Model Development (RF)			■	■				
XGBoost Upgrade				■	■			
Gradio UI Development					■	■		
TTS + Sentence Builder						■	■	
Testing & Debugging							■	■
Deployment & Report							■	■

Table 4.1: Project Gantt Chart (Shaded cells indicate active development weeks)

4.2 Team Contribution

Team Memeber	Responsibility	Contribution%
Raghav Gaur	Dataset collection pipeline, XGBoost model training, MediaPipe integration, predictor.py, overall system architecture, GitHub & HuggingFace deployment	40%
Dev Dixit	Gradio UI development (app.py), sentence builder logic (sentence_builder.py), TTS integration (tts_engine.py), local testing & debugging	30%
Harshita Maggo	Literature review, comparative analysis, dataset validation, project report writing, WBS planning, SDG mapping, presentation preparation	30%

4.3 Work Breakdown Structure

WBS Level 1: Sign Language Recognition System WBS 1.1: Research and Planning

WBS 1.1.1: Literature review of existing sign language recognition approaches WBS 1.1.2: Technology stack selection and feasibility analysis

WBS 1.2: Data Engineering

WBS 1.2.1: Data collection pipeline development (Gesture_Ai_data.py) WBS 1.2.2: Dataset recording (3,316 samples across 5 classes)

WBS 1.2.3: Data validation and preprocessing WBS 1.3: Model Development

WBS 1.3.1: Random Forest baseline implementation WBS 1.3.2: XGBoost upgrade with hyperparameter tuning WBS 1.3.3: Model evaluation and cross-validation

WBS 1.4: Application Development

WBS 1.4.1: MediaPipe integration and feature extraction (predictor.py) WBS 1.4.2: Sentence builder with rolling buffer logic (sentence_builder.py) WBS 1.4.3: Text-to-speech integration (tts_engine.py)

WBS 1.4.4: Gradio UI and streaming interface (app.py) WBS 1.5: Testing and Deployment

WBS 1.5.1: Local testing and performance benchmarking WBS 1.5.2: HuggingFace Spaces deployment

5. Conclusion and Future Directions

5.1 Conclusion

This project successfully demonstrates the development and deployment of a real-time sign language recognition system that is accurate, lightweight, and publicly accessible. By combining Google's MediaPipe Hands framework for efficient hand landmark extraction with an XGBoost classifier for gesture recognition, the system achieves 97.44% test accuracy on a five-class custom dataset while maintaining real-time performance on standard CPU hardware.[2][3][4]

The modular architecture, comprising separate components for prediction, sentence building, text-to- speech, and UI, ensures maintainability and extensibility. The Gradio-based web interface provides an intuitive and accessible demonstration platform, while the HuggingFace Spaces deployment makes the system publicly available without requiring any local installation. The project validates the feasibility of practical, deployable sign language recognition using lightweight computer vision and classical machine learning, providing a strong foundation for further development toward a comprehensive assistive communication tool.[6][15][7]

5.2 Future Directions

- Expand gesture vocabulary to cover the full 26-letter ASL alphabet and common word signs, leveraging publicly available datasets such as the ASL Dataset on Kaggle (27,000+ images) and the WLASL video dataset[14]
- Implement temporal gesture modeling using LSTM or GRU layers on top of MediaPipe landmark sequences to support dynamic (motion-based) signs
- Develop a mobile application using TensorFlow Lite or ONNX Runtime for on-device inference without internet dependency[15]
- Integrate facial expression landmarks from MediaPipe Face Mesh to capture non-manual markers essential for grammatically complete sign language

- Incorporate two-hand gesture support using MediaPipe's multi-hand detection capability
- Explore few-shot learning techniques to allow rapid addition of new gestures from minimal user-recorded samples, personalizing the system to individual signing styles
- Conduct user studies with members of the deaf and hard-of-hearing community to evaluate practical utility and identify usability improvements
- Replace XGBoost with a lightweight CNN + LSTM hybrid, processing raw landmark sequences over time rather than single-frame snapshots, enabling recognition of dynamic signs like "THANK YOU" or "PLEASE" that require motion
- Alternatively, a GRU-based temporal model trained on 30-frame windows per gesture, capturing motion trajectory without heavy compute[16]
- Raspberry Pi 5 or Jetson Nano as edge compute unit with attached wide-angle USB camera for better hand capture range
- Dedicated speaker module for real-time TTS output without browser dependency
- Small HDMI display (7-inch touchscreen) for portable standalone kiosk deployment, removing laptop dependency entirely
- GPU acceleration (NVIDIA RTX or cloud A100) for training on full ASL datasets of 50,000+ samples
- What else we can add:
- Indian Sign Language (ISL) mode — completely separate model, huge scope for India- specific deployment in government offices, hospitals.
- Live video call integration — browser extension that overlays gesture predictions on Zoom/Meet calls with auto-captioning

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